Yes
Yes
Yes
RESOLUTION MSC.35(63)
(adopted on 20 May 1994)
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TOWING ARRANGEMENTS ON TANKERS M6C 63/23/Add.1
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SOURCE: MSC.35(63).pdf
Depending on design
Yes
Yes
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Strength requirements
Yes
Yes
Yes
Yes
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2.3 Strenqth of the towinq components
2.3.1 Towing components as specified in 2.2 for strength should have a working strength of at least 1,000 kN for tankers of 20,000 tonnes deadweight and over but less than 50,000 tonnes deadweight, and at least 2,000 kN for tankers of 50,000 tonnes deadweight "and over (working strength is defined as one half ultimate strength). The strength should be sufficient for all relevant angles of towline, i.e. up to 90
0
from the ship's centreline to port and starboard and
30 0 vertical downwards.
2.3.2 Other components should have a working strength sufficient to withstand the load to which such components may be subjected during the towing operation.
2.4 Lenqth of towinq pennant
The towing pennant should have a length of at least twice the lightest seagoing ballast freeboard at the fairlead plus 50 m.
2.5 Location of stronqpoint and fairlead

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SOURCE: MSC.35(63).pdf
the towing operation.
2.4 Lenqth of towinq pennant
The towing pennant should have a length of at least twice the lightest seagoing ballast freeboard at the fairlead plus 50 m.

2.5 Location of stronqpoint and fairlead
The bow and stern strongpoint and fairleads should be located so as to facilitate towing from either side of the bow or stern and minimize the stress on the towing system.
2.6 Stronqpoint
The inboard end fastening should be a stopper or bracket or other fitting of equivalent strength. The strongpoint can be designed integral with the fairlead.
2.7 Fairleads
2.7.1 ~
2.7.2
2.7.3
Fairleads should have an opening large enough to pass the larqest portion of the chafing gear, towing pennant or towing line.
Geometry
The fairlead should give adequate support for the towing pennant during towing operation which means bending 90
0
to port and to
starboard side and 30
0
vertical downwards. The bending ratio
(towing pennant bearing surface diameter to towing pennant diameter) should be not less than 7 to 1.
Vertical location
The fairlead should be located as close as possible to the deck and, in any case, in such a position that the chafing chain is

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should be not less than 7 to 1.
Vertical location
The fairlead should be located as close as possible to the deck and, in any case, in such a position that the chafing chain is
approximately parallel to the deck when it is under strain between the strongpoint and the fairlead.
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2.8 Chafing chain
Different solutions on design of chafing gear can be used. If a chafing chain is to be used, it should have the following characteristics:
2.8.1
The chafing chain should be stud link chain.
2.8.2 Length
2.8.3
2.8.4
The chafing chain should be long enough to ensure that the towing pennant remains outside the fairlead during the towing operation.

A chain extending from the strongpoint to a point at least 3 m beyond the fairlead should meet this criterion.

Connecting limits

One end of the chafing chain should be suitable for connection to the strongpoint. The other end should be fitted with a standard pear-shaped open link allowing connection to a standard bow shackle.

Stowage

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Connecting limits

One end of the chafing chain should be suitable for connection to the strongpoint. The other end should be fitted with a standard pear-shaped open link allowing connection to a standard bow shackle.

Stowage

The chafing chain should be stowed in such a way that it can be rapidly connected to the strongpoint.

2.9 Towing connection

The towing pennant should have a hard eye-formed termination allowing connection to a standard bow shackle.

2.10 Prototype test

Designs of emergency towing arrangements in accordance with these Guidelines should be prototype tested to the satisfaction of the Administration.

3 READY AVAILABILITY OF TOWING ARRANGEMENTS

3.1 To facilitate approval of such equipment and to ensure rapid deployment, emergency towing arrangements should comply with the following criteria:

.1 The aft emergency towing arrangement should be pre-rigged and be capable of being deployed in a controlled manner in harbour conditions in not more than 15 min .

. 2 The pick-up gear for the aft towing pennant should be designed at least for manual operation by one person taking into account the

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capable of being deployed in a controlled manner in harbour conditions in not more than 15 min .

. 2 The pick-up gear for the aft towing pennant should be designed at least for manual operation by one person taking into account the absence of power and the potential for adverse environmental conditions that may prevail during such emergency towing operations. The pick-up gear should be protected against the weather and other adverse conditions that may prevail .

. 3 The forward emergency towing arrangement should be capable of being deployed in harbour conditions in not more than 1 h.

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.4 The forward emergency towing arrangement should be designed at least with a means of securing a towline to the chafing gear using a suitably positioned pedestal roller to facilitate connection of the towing pennant .
5 Forward emergency towing arrangements which comply with the re'quirements for aft emergency towing arrangements may be accepted .

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SOURCE: MSC.35(63).pdf
suitably positioned pedestal roller to facilitate connection of the towing pennant .
5 Forward emergency towing arrangements which comply with the re'quirements for aft emergency towing arrangements may be accepted .
. 6 All emergency towing arrangements should be clearly marked to facilitate safe and effective use even in darkness and poor visibility.
3.2 All emergency towing components should be inspected by ship personnel at regular intervals and "maintained in good working order.
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